Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/16/2011
PI Name:	Kozlowski, Steve Ph.D.		
Project Title:	Developing, Maintaining, and Restoring Team Cohesion		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	33620	**Congressional District:**	12
Comments:	I moved from Michigan State University to the University of South Florida in August 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	08/16/2009	**End Date:**	08/15/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DeShon, Rick (Michigan State University) Biswas, Subir (Michigan State University) Schmitt, Neal (Michigan State University)		
Grant/Contract No.:	NNX09AK47G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Team cohesion, collaboration, and psycho-social health are essential ingredients for team effectiveness, particularly for those “high reliability” action teams that perform critical tasks under the stress of isolated, confined, and extreme (ICE) environments – such as teams of astronauts. Although there is considerable research that has established a relationship between team cohesion and team effectiveness, there is very little research that examines how teams cohere and collaborate over time. This ground-based research is designed to accomplish three goals: (1) examine interventions to enhance the development and maintenance of team cohesion; (2) create psychological and physiological measurement systems that monitor team psycho-social health and can trigger dynamic interventions to maintain it and restore it; and (3) provide recommendations for training on (a) critical knowledge and skills (i.e., teamwork competencies, relationship management) and (b) monitoring/regulating interventions for psycho-social health maintenance. Project goals will be accomplished using a multidisciplinary team and a combination of research strategies: (1) Laboratory research using a team simulation will be used to (a) examine interventions to enhance team functioning and (b) develop measurement techniques for assessing team collaboration and cohesion. (2) Prototype sensor-based monitoring systems to assess team collaboration and psycho-social functioning. (3) Promising interventions and monitoring/regulation techniques will be examined in the field with long duration teams operating in extreme environments to evaluate proof of concept. The research is designed to resolve IRP Gap BHP 2.3.1 by identifying training content, design, and delivery to build and maintain team cohesion and BHP 2.2.1 developing a monitoring system designed to support, regulate, and repair team cohesion and maintain performance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	At the point of this report, the research is in its infancy, so the benefits are as yet to be realized. However, it is well established in the research literature that team cohesion is related to team effectiveness. Several meta-analytic reviews (i.e., quantitative summaries of the research literature) support this finding, and cohesion is known to be more strongly related to team effectiveness as the team task becomes more complex. Clearly, team cohesion is relevant to the effectiveness of space crews, but how is cohesion relevant to life on Earth? Well, over the last two decades there has been a world-wide restructuring of work systems from individual jobs to teams so as to enable more rapid, flexible, and adaptive decision making. Thus, developing and maintaining team cohesion is not merely relevant to astronaut teams or a range of military teams, it is vital to team effectiveness across a broad cross-section of industry and government. Finally, the monitoring technology that is under development is not just relevant to tracking team cohesion for astronauts, the military, and organizations. It also has the potential to be useful for remotely monitoring the health status, activity levels, and well being of the elderly and individuals who may benefit from remote monitoring.
	Project Organization The project is organized around the 3 primary research foci: • The Simulation Team (Optima Lab) is tasked with developing team simulations, constructing interventions to promote team collaboration; adapting measurement techniques to assess cohesion, collaboration, and interaction over time; and conducting research to evaluate the effectiveness of interventions. o Key personnel include Dr. Steve Kozlowski, PI; Dr. Rick DeShon, Co-I; and research assistants Samantha Baard, Mike Braun, Marina Jackson (funded), and Tara Rench (funded). • The Monitoring Technology Team is tasked with developing hardware and software to enable wireless monitoring of team member collaboration and engagement o Key personnel include Dr. Subir Biswas, Co-I; and research assistants Yogesh Piolet (funded) and Dong Bo. o Validation research and integration with psychological assessments of team collaboration and effectiveness is being conducted by the simulation team in conjunction with the monitoring team. • The Analog Team is tasked with developing access to analog teams (i.e., long duration, isolated, extreme conditions) to collect data and conducting benchmarking research to examine relationships between individual differences, team cohesion, and team effectiveness. o Key personnel for the original effort included Neal Schmitt, Leaetta Hough, and James Locke. However, problems gaining access to the analog environment have put this effort on indefinite hold. o An alternative team led by the PI has assumed responsibility for this research focus. Key personnel include Dr. Steve Kozlowski, PI; Dr. Rick DeShon, Co-I; and research assistants Samantha Baard, Mike Braun, Marina Jackson, and Tara Rench. Research Progress Simulation Research Team • Task 1-1: Benchmark Cohesion Formation (completed). • Task 1-2: Revise Dynamic Team Simulation (completed). • Task 1-3: Identify Individual Differences Relevant to Team Composition (completed). • Task 1-4: Develop Interventions (in progress). A teamwork skills training intervention has been developed. Current experimentation is focused on developing and refining behavioral metrics to assess team cohesion / collaboration. • Task 1-5: Develop Psychological States Assessment Techniques (completed).

Task Progress:

• Task 1-6: Conduct Team Simulation Research (in progress). Two studies were conducted.

o Study 1 examined familiar and unfamiliar teams who worked on a complex team simulation once a week across a five week period. This effort was designed to assess fluctuations in team cohesion over time and its relationship to team performance. Self-report measures of cohesion and other team process indicators were highly intercorrelated. Results indicated that unfamiliar teams experienced significantly more variability in levels of cohesion relative to familiar teams and that cohesion was more tightly coupled to performance for unfamiliar teams. These findings accentuate the importance of examining cohesion and collaboration dynamically.

o Study 2 examined 3-person teams performing MARSim across 10 scenarios. Given the challenges of studying cohesion using self-reports (i.e., high intercorrelations among self-report indicators), this research pioneered behavior-based assessments of team collaboration. Results indicated that team cohesion (self-reports) accounted for 5% of team performance variance above and beyond experience (trial at 27% variance accounted). Importantly, the behavioral indicators of collaboration accounted for an additional 17% of performance variance beyond experience and self-reported cohesion. These findings provide strong preliminary evidence for the viability of assessing team cohesion / collaboration behaviorally.

Monitoring Technology Team

• Task 2-1: Design and Development of the Wearable Wireless Network (Year 1). The objectives for the first year of the project included (a) network design and system development, (b) network protocol development, and (c) data processing algorithm development.

• Task 2-1: Design and Development of the Wearable Wireless Network (Year 2). New packaging was developed for the wearable sensor.

1. Development of audio processing algorithms for speaker detection using wearable sensors. Specific progress includes:

a. Several laboratory experiments were carried out for collecting audio data from two wearable sensors while two individuals are conversing.

b. A number of parameters (i.e. Distance between the speakers, Loudness of their individual voice, Different audio sampling rates) were varied during those experiments.

c. An algorithm was developed that can be used for processing data within the wearable sensors so that for a conversation the sensor can detect which participant spoke for what duration.

d. An offline version of the algorithm was also developed, so that after all data is collected in a PC, the detection can be done in the PC.

• Task 2-2: Laboratory Validation of Sensor Data and Wireless Network (in progress).

1. Preliminary validation of the wearable sensor network is in progress. A team of graduate students from Psychology and Engineering collected team (three members) interaction data using the wearable sensors as well as digital video recorders. The following graph shows face time statistics among the members A, B, and C during a 30-min session of collaborative interaction session. Video validation of this result is currently being conducted.

2. Preliminary validation findings are supportive of the monitoring technology as a source of metrics that assess team collaboration. Validation continues.

Analog Teams

• Task 3-1: Conduct Antarctic Field Research. This research is technically not scheduled until year 3 of the project. However, we believed that it would be useful to collect individual differences (i.e., personality), team process, and team outcome data from analog teams operating in ICE environments as part of our effort to benchmark cohesion formation. Our project began too late in 2009 to be able to conduct a prospective study, but we received IRB approval (NASA for NSF-OPP; MSU IRB) to conduct a retrospective study of teams in the most recent deployment. The intent was to use this initial effort to build relationships, gain access, and get experience in the research setting that would aid our latter efforts. This effort is currently stalled as NSF is apparently reluctant to facilitate access. James Locke, Leaetta Hough, and Neal Schmitt are / were leading this effort.

• Given the difficulties highlighted above, this year Kozlowski and DeShon explored alternative analog team options with Dr. Keeton (ANSMET, Concordia Station, NEEMO) that could provide benchmarking data (now) and opportunities to evaluate interventions and monitoring (later). We were fortunate to be able to gain access to the Antarctic Search for Meteorites (ANSMET) team for their 2010-2011 deployment to the Antarctic. Although the sample size was somewhat small, the ability to examine team members working in close collaboration for an extended period time (6 weeks) under ICE conditions was very informative. Much like the findings for the benchmarking sample and the simulation teams, cohesion varied considerably over the duration of the ANSMET mission. In particular, there were substantial differences in cohesion variability across the two different types of teams: recon and systematic (search). As shown in Figure 5, there was much more change in cohesion over time in the systematic team versus the recon team. Specifically, while the recon team's cohesion stayed relatively consistent throughout the duration of the six-week mission, the systematic team's cohesion was more volatile. We identified factors that influenced cohesion, including positive triggers like engaging in fun activities together and being productive as a team; and negative triggers like stress/fatigue, bad weather conditions, and uneven distribution of work among team members. Those factors were associated with peaks and valleys of team cohesion.

Research Summary

A substantial amount of cross-sectional (i.e., static) research has shown a strong relationship between team cohesion and team performance. Although team cohesion is generally assumed to be a relatively stable team process variable, little is known about its potential dynamics. Across benchmarking data (military teams during 8 weeks of intensive training), simulation teams (across hours to weeks of performance), problem solving teams (hours), and analog teams (6 weeks deployed in an ICE environment), we observed considerable variation in team cohesion and team collaboration effectiveness. This convergent evidence across very different teams, tasks, and timeframes makes it evident that the assumed stable relationship between team cohesion and team effectiveness needs to be examined closely and unpacked.

	This research is developing metrics, tools, and techniques to do just that. Schedule Status Key tasks are on schedule and the project is making progress. There have been some delays, however, that should be noted. Initial project start-up was delayed by administrative difficulties within Contracts and Grants Administration (CGA) at MSU. Essentially, it took 2 months for CGA to get an account set up; this delay in the provision of funds for the project slowed somewhat our initial progress. In addition, there two areas that merit discussion. First, we have refocused our simulation research to concentrate on behavioral assessments of cohesion and collaboration. This is important, in our view, for developing a suite of assessments to complement the monitoring technology. However, this focus on behavioral assessment has delayed our evaluation of training interventions. Second, access to analog environments for both (a) baseline assessments of team functioning over long duration in ICE environments and (b) evaluation of monitoring technology / metrics continues to be a challenge. We are very grateful for the assistance that Dr. Keeton provided this year to connect us to ANSMET. Nonetheless, this is an issue of continuing concern.
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